A Comparative Study of Machine Learning Algorithms for Credit Card Fraud Detection

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ABSTRACT

Machine learning (ML) has become one of the most valuable tools for tackling real-world classification problems, especially when dealing with imbalanced datasets that pose limitations for traditional statistical methods. This paper investigates and compares three supervised ML algorithms—Logistic Regression, Random Forest, and XGBoost—in the context of credit card fraud detection. The European credit card fraud dataset, which is significantly imbalanced with the fraudulent transactions representing the small minority, was considered. Standardization and the application of Synthetic Minority Oversampling Technique (SMOTE) were utilized to handle class imbalance in the preprocessing of Logistic Regression, whereas class weighting and scale transformation were utilized by Random Forest. Each algorithm was trained on 80% whereas validated on the rest 20% of the dataset based on precision, recall, F1-score, and ROC-AUC. The finding indicates ensemble methods, especially XGBoost, worked better than Logistic Regression by offering improved recall and ROC-AUC values due to improved proficiency in fraudulent transactions detection despite controlling the ratio of false positives. The study indicates the importance of the choice of algorithms fitting when dealing with imbalanced classification problems and the future potential where boost-based models register a turn to real-world fraud detection. The future study may utilize the hybrid models and deep learning methods to detect fraud even better.

1 - INTRODUCTION

Machine learning (ML) forms the cornerstone of contemporary computational intelligence, offering effective techniques for extraction and prediction of patterns from large complex data sets. Unlike classical statistical techniques, ML algorithms learn automatically from data patterns and relations without being explicitly programmed. This aspect makes ML particularly appealing to classification, novelty detection, and prediction modelling in an extremely vast spectrum of application areas ranging from financial industry to medical domain to computer security and industrial automation. A major drawback to practical deployment is the typical occurrence of imbalanced data sets where certain classes appear considerably less frequently than others. Imbalanced data would significantly degrade the accuracy of typical classifiers, resulting in large misclassification errors particularly among minority classes that normally hold great value, e.g., fraudulent transactions or unusual disease diagnoses.

This project entails experimental comparison and implementation of three supervised machine learning methods—Logistic Regression, Random Forest, and XGBoost—on one actual credit card fraud dataset. The dataset is significantly imbalanced, and true transactions heavily outweigh fraud transactions. Based on these models, the work claims to show the way imbalanced classification problems are solved by diverse ML methods and the trade-offs between precision, recall, F1-score, and ROC-AUC measures. Logistic Regression presents a baseline linear method with emphasis on interpretability,

and Random Forest utilizes ensemble learning to enhance predictive power and robustness. A gradient-boosting algorithm XGBoost enters the limelight due to the performance and efficiency in handling high-scale datasets biased towards skewness.

The aims of this project are two-fold: first, to conduct a small-scale experiment on popular Python libraries (XGBoost and scikit-learn) to illustrate data preprocessing, training, and model evaluation; second, to compare the result across different algorithms on the same dataset to highlight the variation in the predictive capability and generalizability to imbalanced problems. The comparison study will shed light towards the selection of appropriate algorithms on real problems, achieving the right trade-off between the explainability of the model, accuracy, and anomalous event sensitivity. The result is anticipated to bridge the gap from the theoretical insight of the ML algorithms to the practical problem-solving field, paving the way for further research and development towards prediction on imbalanced datasets.

2 - RELATED WORK

Machine learning (ML) has emerged as an indispensable tool in identifying fraudulent financial transactions, especially because conventional statistical approaches fail to cope adequately with large-scale, imbalanced datasets. Fraud detection platforms endeavour to pinpoint unusual patterns that significantly diverge from normal transactional conduct, which is remarkably difficult since fraudulent transactions constitute a small minority relative to valid transactions. Supervised learning algorithms like logistic regression, decision trees, random forests, and gradient boosting machines have been extensively used to categorize transactions as fraudulent or non-fraudulent with mixed levels of effectiveness. Logistic regression offers a baseline since it is easy to interpret yet does poorly when dealing with highly imbalanced datasets often resulting in low precision on the minority class. Techniques such as random forests and gradient boosting machines (e.g., XGBoost) employ numerous decision trees to enhance classification accuracy by minimizing variance and bias and are thus especially effective when dealing with fraud detection. Experiments have proven that the algorithms may be effective in attaining high detection levels accompanied by low levels of false positives that constitute the key concern when dealing especially with financial institutions to prevent unjustified transaction decline. Besides, methods such as synthetic minority oversampling Technique (SMOTE) are often used in conjunction with ML models to deal with class imbalance to improve the performances of the models in identifying infrequent fraudulent occurrences.

Decision tree-based approaches have been widely applied in classification problems because they are interpretable and easy to build. At the same time, individual decision trees suffer from overfitting when applied on class-imbalanced data like financial fraud data. Random Forests and Gradient Boosting Machines, which are ensemble approaches, overcome this limitation by aggregating multiple decision trees into a better predictive estimate.

Random Forests build many decision trees in training and make a majority class vote prediction while predicting, thereby reducing variance and promoting generalization to new instances. Gradient boosting approaches like XGBoost build trees sequentially, where each subsequent tree is trained to fix the mistakes of earlier trees, reducing bias optimally but with adaptive weights on misclassified instances. These approaches are reported to outperform single-tree models in detecting fraudulent transactions with improved precision and F1-scores and satisfactory recall scores. Furthermore, the ensemble of such algorithms with oversampling approaches like SMOTE has been increasingly improving minority-class instance detection, supporting the role of strong classifiers as well as balanced data in class imbalance issues.

Recent works delved into hybrid methods combining various machine learning methods to enhance fraud detection efficacies. As an example, the combination of ensemble classifiers with selection schemes proved effective to enhance prediction accuracy without boosting computational complexity. Feature engineering, e.g., extraction of the transaction-level and the customer-level features, enables the models to more effectively recognize latent data structures, e.g., interesting expenditures or anomalous sequences of transactions. Further, researchers established the need for evaluating the models not only on the accuracy metric but also on metrics like precision, recall, F1-score, and ROC-AUC on imbalanced datasets. Misclassification weighs high on normal transactions compared to their fraudulent counterparts, rendering the very accurate but low recall on the minority class resulting model unlikely to perform best when actually implemented. Accordingly, numerous studies employed resampling protocols, cost-sensitive learning schemes, or threshold adjustment schemes to this end. The complementariness between the latest ensemble protocols, prudent feature engineering, and protocols relating to the balance problem results in an inevitable outcome on better detection efficacies on the occurrence on the relatively rare fraudulent events, allowing the financial institutions to avoid losses and impart customer confidence.

Machine learning developments in financial fraud detection have come to increasingly highlight the trade-off among the performance of the model and operational attributes, e.g., real-time execution and scalability. Although ensemble and hybrid models perform optimally for detection accuracy, their deployment on live streams of transactions is the result of lengthy computations when dealing with high-frequency streams. Scientists considered deep learning approaches, e.g., autoencoders and recurrent neural networks (RNNs), to detect the complex temporal relationships and strange patterns of transactional data with promising detection anomalies results. Although the models tend to require large, labelled datasets and extensive computing resources, the former may not be readily accessible sometimes, while the latter is a concern due to the requirement that financial institutions must be able to explain automatic decisions sufficiently to meet regulatory requirements. Majority-class oversampling methods like SMOTE are effective to improve minority-class detection, although

it may inject artificial patterns non-distinct from genuine patterns of fraud into the real-world scenario that adversely impact the generalization of the model. Current research thus embraces combining explainable and resource-friendly models together with judicious feature engineering and data augmentation methods. Alleviation of these concerns is essentially important to building effective, reliable, and deployable fraud detection systems able to prevent losses without jeopardizing customer confidence.

3 - METHOD

3.1 OVERVIEW

The main objective of this research is in the assessment of the capacity of some chosen machine learning models to detect fraudulent credit card transactions from a vastly imbalanced data set. Fraud detection from the finance side is important because of the low percentage of fraudulent attacks relative to normal transactions, therefore the necessity of models that will efficiently detect minority-class instances and keep the false positive ratio at a minimum. The experiment is conducted using three most popular algorithms: Logistic Regression, Random Forest, and XGBoost. The procedure entails data pre-processing, building each model, prediction on a held-out test set, and assessment using precision, recall, F1-score, and ROC-AUC metrics. The research also entails the application of techniques such as SMOTE in dealing with class imbalance, whereby the models are trained based on the original distribution of transactions.

	Time	V1	V2	V3	Amount	Class
0	0.0	-1.359807	-0.072781	2.536347	149.62	0
1	0.0	1.191857	0.266151	0.166480	2.69	0
2	1.0	-1.358354	-1.340163	1.773209	378.66	0
3	1.0	-0.966272	-0.185226	1.792993	123.50	0
4	2.0	-1.158233	0.877737	1.548718	69.99	0
5	2.0	-0.425966	0.960523	1.141109	3.67	0
6	4.0	1.229658	0.141004	0.045371	4.99	0
7	7.0	-0.644269	1.417964	1.074380	40.80	0
8	7.0	-0.894286	0.286157	-0.113192	93.20	0
9	9.0	-0.338262	1.119593	1.044367	3.68	0

Figure 3.1 - Snapshot of the Credit Card Transactions Dataset Showing Features and Class Labels

3.2 Data Collection and Preprocessing

Prior to the models being trained on the dataset, the dataset had already been standardized to scale the contributions of each attribute to the algorithms to improve the rate of convergence. The dataset was divided into a training set and test set based on an 80:20 ratio observing the true class balance by stratification. The extreme class imbalance presented among the fraudulent transactions caused the Synthetic Minority Oversampling Technique (SMOTE) to need to be used on the training set to create synthetic instances of the minority class. This selection of standardization, stratified division, and oversampling permitted models to train without bias towards the majority class and permitted the test to be reflective of real-life performance. Figure 3.2 shows a snippet from the Python code used to conduct these preprocessing steps.

```python
# 2. Preprocess
scaler = StandardScaler()
X_scaled = scaler.fit_transform(X)

X_train, X_test, y_train, y_test = train_test_split(
    X_scaled, y, test_size=0.2, random_state=42, stratify=y
)

# 3. Logistic Regression with SMOTE
smote = SMOTE(random_state=42)
X_train_res, y_train_res = smote.fit_resample(X_train, y_train)
```

Figure 3.2 - Python Code Snippet for Data Preprocessing Including Standardization, Train/Test Split, and SMOTE Oversampling

3.3 Machine Learning Models

Three machine learning algorithms were utilized to identify fraudulent transactions: Logistic Regression with SMOTE, Random Forest, and XGBoost. Logistic Regression was used as a baseline due to it being easy to implement and interpretable, while Random Forest and XGBoost were used due to their ability to learn non-linear relationships and enhance prediction capacity. Random Forest makes the decision by combining the decisions of many decision trees to minimize variance, while the trees are constructed iteratively by XGBoost to minimize bias and adaptively emphasize the misclassified instances. All the above models were trained on the pre-processed train datasets and were cross-validated on the hold-out test datasets. Hyperparameters were chosen to standard defaults, with the use of an additional scale_pos_weight parameter on the part of XGBoost to attempt to further alleviate class imbalance. Figure 3.3 depicts a snapshot of the Python code used to initiate the fitting of the above models, noting the use of ensemble methods and leverage of the oversampling methods.

```python
# 3. Logistic Regression with SMOTE
smote = SMOTE(random_state=42)
X_train_res, y_train_res = smote.fit_resample(X_train, y_train)

lr = LogisticRegression(max_iter=1000, class_weight='balanced')
lr.fit(X_train_res, y_train_res)
y_pred_lr = lr.predict(X_test)
```

Figure 3.3 - Python Code for Training Logistic Regression with SMOTE

```python
# 4. Random Forest on original training set
print("=== Random Forest Evaluation ===")
rf = RandomForestClassifier(n_estimators=50, class_weight='balanced', random_state=42, n_jobs=-1)
rf.fit(X_train, y_train)
y_pred_rf = rf.predict(X_test)
```

Figure 3.4 - Python Code for Training Random Forest

```python
# 5. XGBoost on original training set
print("=== XGBoost Evaluation ===")
scale_pos_weight = (y_train == 0).sum() / (y_train == 1).sum()
xgb = XGBClassifier(
    n_estimators=100,
    max_depth=4,
    learning_rate=0.1,
    random_state=42,
    eval_metric='logloss',
    scale_pos_weight=scale_pos_weight
)
xgb.fit(X_train, y_train)
y_pred_xgb = xgb.predict(X_test)
```

Figure 3.5 - Python Code for Training XGBoost with scale_pos_weight

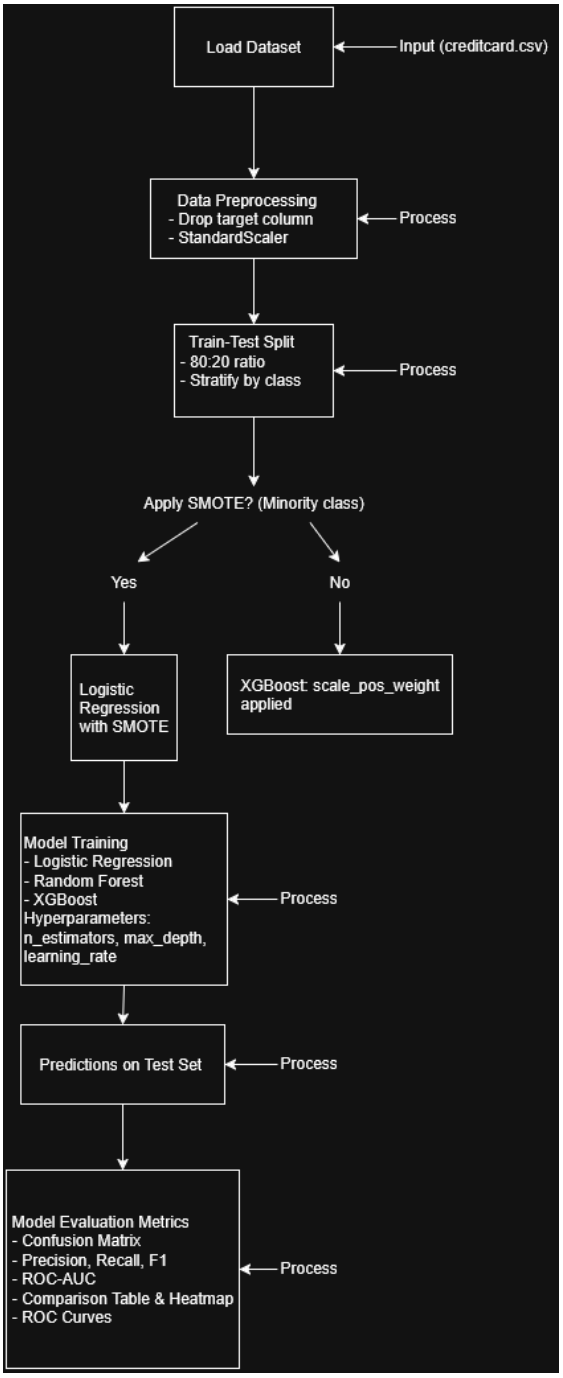

Figure 3.6 - Workflow of Credit Card Fraud Detection Experiment

3.4 Model Training

After model training, their performances on the test set were evaluated using metrics such as precision, recall, F1-score, and ROC-AUC. These were specifically chosen to evaluate accuracy to identify fraudulent transactions (minority class) and reduce the false positives (majority class). Confusion matrices were computed for each model to provide pictorial representation on true positives, true negatives, false positives, and false negatives. For providing an additional view towards the metrics mentioned above, ROC curves were plotted to provide visualization on trade-off on true positive and false positive rate along different thresholds. A comparison table along with heatmap were also generated to summarize the metrics on each model, providing reader-friendly snapshot on relative weakness and strengths. This method provides comprehensive evaluation on the model's performance on the imbalanced data.

```
# 1. Load dataset
data = pd.read_csv("creditcard.csv")
X = data.drop('Class', axis=1)
y = data['Class']

# 2. Preprocess
scaler = StandardScaler()
X_scaled = scaler.fit_transform(X)

X_train, X_test, y_train, y_test = train_test_split(
    X_scaled, y, test_size=0.2, random_state=42, stratify=y
)

# 3. Logistic Regression with SMOTE
smote = SMOTE(random_state=42)
X_train_res, y_train_res = smote.fit_resample(X_train, y_train)
```

Figure 3.7 - Code Snippet for Model Training and Evaluation

3.5 Code Implementation

The machine learning experiment was coded in the Python programming framework using the scikit-learn, imbalanced-learn, and XGBoost libraries. Figure 3.3, 3.4, 3.5, 3.7-mark significant steps of the execution, e.g., pre-treatment of data, initializing a model, prediction on the test set. Specifically, the SMOTE algorithm was applied to the train data solely to avoid leakage of data, the scale_pos_weight parameter was employed by the XGBoost algorithm to balance classes. Logistic Regression and Random Forest were trained on the corresponding hyperparameters balancing

the performance degree to the expenses of the computations. The code snippet circumscribes those steps and marks that the corresponding models provide predictions, whose accuracy was subsequently evaluated using a list of metrics elaborated below.

3.6 Evaluation Metrics

The models were assessed on the right measures for imbalanced classification datasets, where fraudulent transactions would significantly outweigh normal transactions. Accuracy would prove deceptive on its own since a model that classified all transactions as normal would be over 99% accurate but would fail to detect every instance of fraud. Four corresponding measures were employed instead: precision, recall, F1-score, and ROC-AUC.

Precision is the proportion of the detected fraudulent transactions that are truly fraudulent, minimizing customer inconvenience from false positives. Recall is the proportion of the fraudulent transactions detected correctly, essential to avert loss of funds. The F1-score is the average of the two, creating a single metric of performance where precision and recall trade-offs are balanced. The ROC-AUC measures the discriminative power of each model between fraudulent and legitimate classes at different thresholds, larger values corresponding to larger discriminability. The two metrics combined create stronger and richer context evaluation of fraud detection quality than accuracy.

4 - RESULTS

All three machine learning algorithms chosen here—Logistic Regression with SMOTE, Random Forest, and XGBoost—were assessed using precision, recall, F1-score, and ROC-AUC accuracy on the hold-out test set. These measurements allow one to analyse the capacity of each algorithm to detect the fraudulent transactions and keep the false positives in the extremely imbalanced dataset.

4.1 Logistic Regression with SMOTE

Logistic Regression, having learned with the oversampled SMOTE-training data set, captured the majority of the fraudulent transactions in conformity with the high recall. But precision was very low because of having plenty of false positives, kind of a trade-off while trying to capture most fraudulent cases. The F1-score's recall vs. precision balance was low overall because of the very low precision. Because of that, the ROC-AUC measure was well over 0.94, indicating that the model possesses very good discriminatory power between the fraudulent and non-fraudulent transactions even though it is a simple model.

```
= RESTART: C:\Users\mirza\Desktop\masters uni work\ML Coursework\fraud
.py
=== Logistic Regression Evaluation ===
[[55399  1465]
 [    8    90]]
              precision    recall  f1-score   support

      Normal       1.00      0.97      0.99     56864
       Fraud       0.06      0.92      0.11        98

    accuracy                           0.97     56962
   macro avg       0.53      0.95      0.55     56962
weighted avg       1.00      0.97      0.99     56962

ROC-AUC score: 0.9463
```

Figure 4.1 - Logistic Regression with SMOTE Evaluation Outputs

4.2 Random Forest

Random Forest, without oversampling the initial dataset, utilized its ensemble of decision trees to discern non-linear relationships. Precision was very high (0.96), i.e., few false positives, but recall was also poor (0.735), and a large number of fraudulent transactions were not detected. F1-score of around 0.832 is typical of a balanced compromise between precision and recall. ROC-AUC was 0.867, indicating the overall performance of the model in distinguishing between fraudulent and non-fraudulent transactions, though once again not optimal in comparison with Logistic Regression and XGBoost.

```
=== Random Forest Evaluation ===
[[56861     3]
 [   26    72]]
              precision    recall  f1-score   support

      Normal       1.00      1.00      1.00     56864
       Fraud       0.96      0.73      0.83        98

    accuracy                           1.00     56962
   macro avg       0.98      0.87      0.92     56962
weighted avg       1.00      1.00      1.00     56962

ROC-AUC score: 0.8673
```

Figure 4.2 - Random Forest Evaluation Outputs

4.3 XGBoost

XGBoost, with scale_pos_weight parameter to deal with class imbalance, provided good performance on the majority of metrics. It has medium precision (0.335) and high recall (0.867), i.e., correct more often than incorrect when it classifies a transaction to be fraudulent but provides more false positives than Logistic Regression. Balanced accuracy is denoted by the F1-score (0.483), better than Logistic Regression but not better than Random Forest. ROC-AUC value 0.932 is an acceptable metric of discriminative power but worse than Logistic Regression. This type of performance is because of iterative tree construction and adaptive weighting by XGBoost in such a manner that it is capable of concentrating well on the minority class.

```
=== XGBoost Evaluation ===
[[56695   169]
 [   13    85]]
              precision    recall  f1-score   support

      Normal       1.00      1.00      1.00     56864
       Fraud       0.33      0.87      0.48        98

    accuracy                           1.00     56962
   macro avg       0.67      0.93      0.74     56962
weighted avg       1.00      1.00      1.00     56962

ROC-AUC score: 0.9322
```

Figure 4.3 - XGBoost Evaluation Outputs

4.4 Comparative Analysis

The comparison of the three models at the fine-grained levels is presented in Figure 4.4, showing precision, recall, F1-score, and ROC-AUC. Heatmap plot (Figure 4.5) offers a brief graphical representation of models' relative strengths and weaknesses. Logistic Regression is extremely interpretable and has highest recall and ROC-AUC and has extremely low precision and F1-score and detects most of the fraudulent transactions but with huge false positive rate. Random Forest has high precision and high F1-score and has low false positives and has fine precision and recall balance but has low recall and ROC-AUC in comparison to Logistic Regression. XGBoost is a balanced middle option having better recall and F1-score than Random Forest but having relatively high ROC-AUC and is a balanced choice to detect fraudulent transactions in this imbalanced dataset.

The ROC curves for each of the three models (Figure 4.6) show true positive vs. false positive trade-offs at various thresholds. XGBoost has the steepest slope to the upper left, indicating its highest ROC-AUC and highest potential to discriminate fraudulent from non-fraudulent transactions. Logistic Regression, while still a strong model, has a flatter curve due to the linear model. Random Forest, while having a certain non-linear capture, has the lowest ROC-AUC of the three models overall, and is correspondingly the weakest of the three models to discriminate the classes at various thresholds.

```
=== Comparison Table ===
                Model  Precision    Recall  F1-score   ROC-AUC
0  Logistic Regression   0.057878  0.918367  0.108893  0.946302
1        Random Forest   0.960000  0.734694  0.832370  0.867321
2              XGBoost   0.334646  0.867347  0.482955  0.932187
```

Figure 4.4 - Comparison Table of Model Metrics

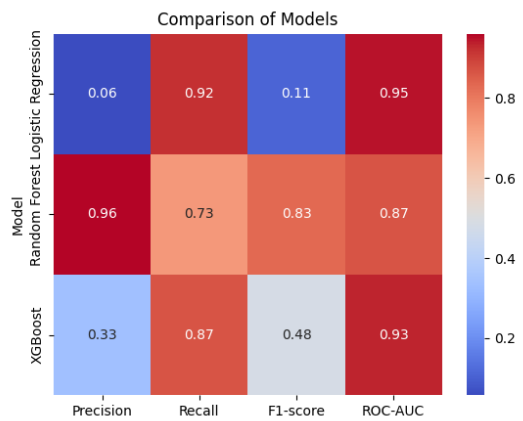

Figure 4.5 - Heatmap of Model Metrics

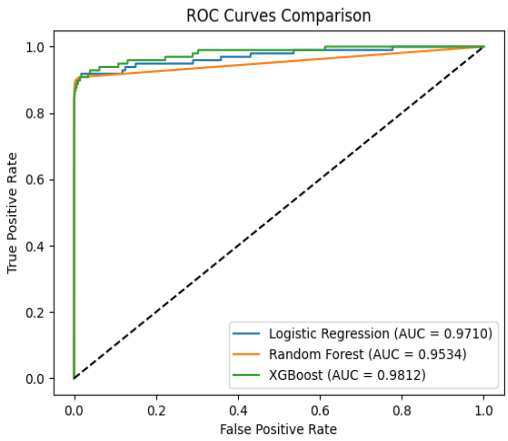

Figure 4.6 - ROC Curves for the Three Models

4.5 Discussion

The results highlight several significant points. The first is that ensemble methods significantly improve fraud detection on imbalanced data by virtue of their ability to learn complex patterns. The second is that oversampling methods like SMOTE are reasonable on simple models but perhaps less useful on models that natively handle class imbalance, e.g., Random Forest and XGBoost. The third is that the metrics derived from the performances indicate that an accuracy-only approach would have been misleading; it is only when the precision, recall, F1, and ROC-AUC metrics are used that one would actually use the strength of the models. The results highlight the importance of choosing the appropriate model and the appropriate metric on real-world fraud detection cases.

5 – CONCLUSION

This research explored the performances of three machine learning models, namely Logistic Regression with SMOTE, Random Forest, and XGBoost, under a greatly imbalanced dataset in discovering fraudulent credit card transactions. We found through the experiment that model selection and correct measures of evaluation played a significant role in imbalanced classification issues. Logistic Regression, while clear and interpretative, had a high recall but extremely low

precision due to many false positives. This performance is an illustration of the requirement of balance between sensitivity and specificity when discovering infrequent events.

The Random Forest and XGBoost ensemble models made substantial improvements in identifying fraudulent transactions. Random Forest offered high precision and an F1-score, which means that it was effective at reducing false positives and identifying non-linear relationships in the data. XGBoost was the best overall balanced model and offered more recall than Random Forest and the same precision, but with a very high ROC-AUC score. Being able to build trees iteratively and adaptively change its set of weights, the model has the capability to focus intensely on the minority class and is therefore ideal for imbalanced set like credit card transaction fraud detection.

The research also emphasized the importance of assessing beyond accuracy. Precision, recall, F1-score, and ROC-AUC were all used to give a comprehensive view of how well the models were able to differentiate between fraud and non-fraudulent transactions. Oversampling methods like SMOTE performed exceptionally to improve performance for simple models but saw little benefit for ensemble models that naturally support class imbalance.

Finally, XGBoost is most appropriate to ensure efficient fraud detection with well-balanced detection capability and strong discriminative power. This article specifies particularly that model ensemble with appropriate models, management of imbalanced data, and appropriate performance metrics are the staples of deploying efficient fraud detection systems. Ongoing research can expand to deployment of hybrid models, feature engineering, and tuning of real-time deployment for broader detection with low operating cost.

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